

1 Problem Setup

$X_1, \dots, X_n$ represents the fiber in the butternut group, and $Y_1, \dots, Y_n$ in the acorns.

We assume the subjects are normally distributed around our measure of interest (with the same variance). That is,

$$X \sim N(\mu_1, \sigma^2)$$

$$Y \sim N(\mu_2, \sigma^2)$$

The goal is to determine if acorns have a higher average fiber than butternuts.

2 C calculation

To accomplish this, we must first define “ α ”:

α represents the probability of the observed difference in sample means exceeding a specific value C , assuming that there truly is no difference in the mean fiber of acorns and butternuts. That is,

$$\alpha = P(\bar{Y}_n - \bar{X}_n > C \mid \mu_2 = \mu_1)$$

Note that C depends on the predetermined α -level. The remainder of this section focuses on calculating C .

Simplifying this probability requires us to utilize distributional properties of our sample means.

Note that the variance of a sample mean equals σ^2/n , and hence the variance of our difference is twice that:

$$\bar{Y}_n - \bar{X}_n \sim N(\mu_2 - \mu_1, 2\sigma^2/n)$$

Under the null hypothesis that $\mu_2 = \mu_1$, we have

$$\bar{Y}_n - \bar{X}_n \sim N(0, 2\sigma^2/n)$$

Probabilities under the standard normal distribution $N(0, 1)$ have nice properties which make evaluation simpler. The next step in evaluating our probability is to convert it into terms of the standard normal.

Clarifying Example

Scaling a variable, which follows a normal distribution, by a quantity of σ multiplies its variance by σ^2 .

For example, let $Z \sim N(0, 1)$. Then,

$$\sigma Z \sim N(0, \sigma^2)$$

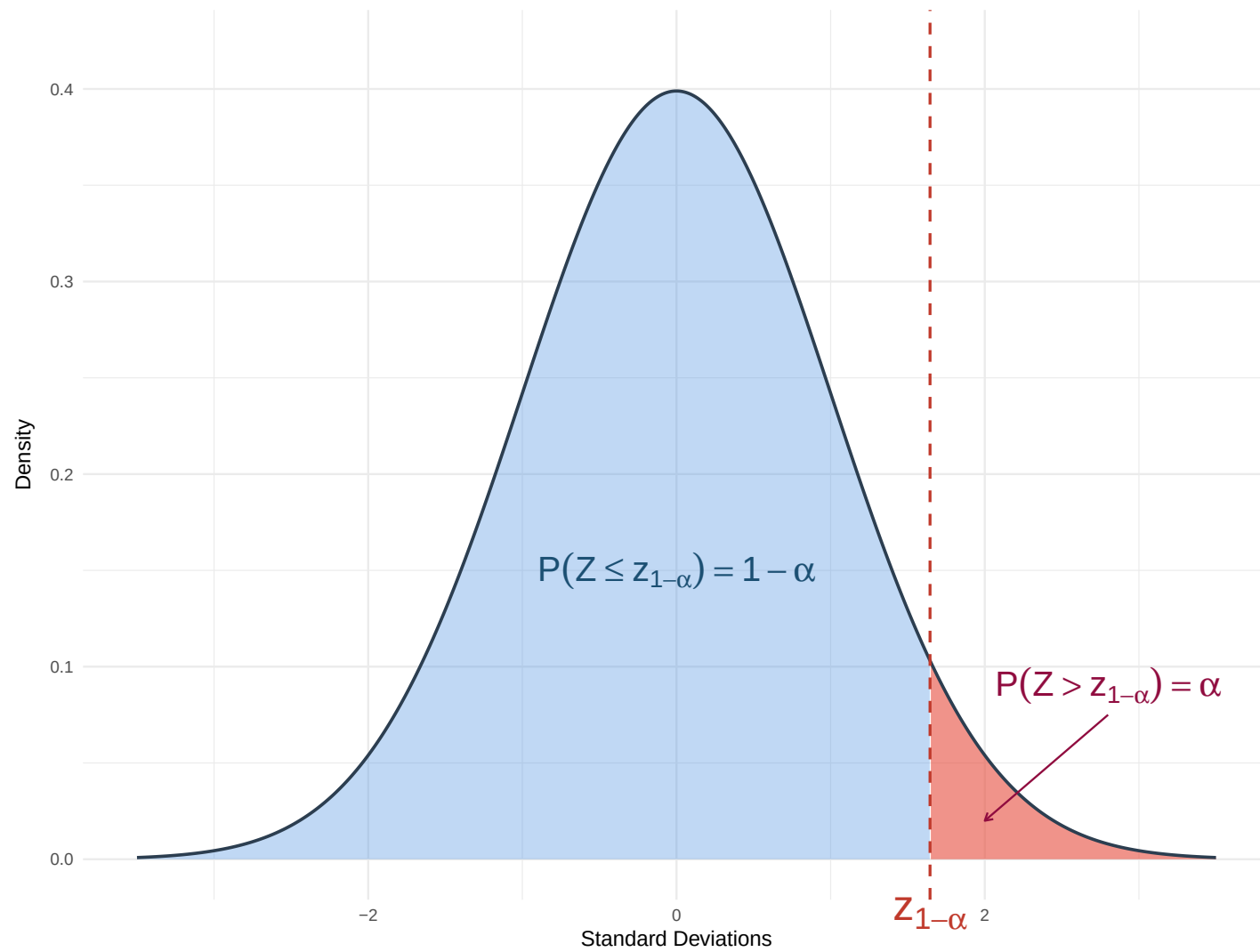
Utilizing this property:

$$(\bar{Y}_n - \bar{X}_n) \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}} \sim N(0, 1)$$

Now, our probability becomes

$$P(\bar{Y}_n - \bar{X}_n > C \mid \mu_2 = \mu_1) = P\left\{(\bar{Y}_n - \bar{X}_n) \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}} > C \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}}\right\}$$

We want the RHS of the inequality inside our probability to represent the $1 - \alpha$ quantile for the probability to be equal to α . *(see figure)*



Notationally, because $LHS = (\bar{Y}_n - \bar{X}_n) \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}} \sim N(0, 1)$, we have

$$RHS = z_{1-\alpha} = C \cdot \frac{\sqrt{n}}{\sigma\sqrt{2}}$$

$$\implies C = z_{1-\alpha} \cdot \frac{\sigma\sqrt{2}}{\sqrt{n}}$$

That is,

$$\alpha = P\left(\bar{Y}_n - \bar{X}_n > z_{1-\alpha} \cdot \frac{\sigma\sqrt{2}}{\sqrt{n}} \mid \mu_2 = \mu_1\right)$$

If we want there to be a 5% chance of...

3 Sample size calculation

Power $(1 - \beta)$ is defined as the probability of rejecting the idea that butternuts and acorns have the same amount of fiber, when in reality acorns have more. I.e.,

$$1 - \beta = P \left(\bar{Y}_n - \bar{X}_n > z_{1-\alpha} \cdot \frac{\sigma\sqrt{2}}{\sqrt{n}} \mid \mu_2 > \mu_1 \right)$$

We want our sample size n to be large enough...